DS603 PROBABILITY FOR DATA SCIENCE · GROUP PROJECT

Tail Risk in Clean vs Traditional Energy Equities

Normal vs Student’s t · Markov and Chebyshev bounds · Monte Carlo LLN and CLT

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ICLN vs IXC · daily returns 2021–2025 · AY2026/27 August Term

RESEARCH QUESTION

Do clean-energy and traditional-energy returns differ in tail risk, and how well do Normal and t models, Markov and Chebyshev bounds, and the sample mean describe it?

Capital is shifting

Investors moving from oil and gas into renewables may be taking on a different kind of extreme-day risk.

Different shocks

Clean energy reacts to policy and interest rates; traditional energy to oil-price shocks.

Tools assume a shape

Many risk tools assume Normal returns. We test when that is safe, and what to use instead.

DATA

Two global energy ETFs, five years of daily returns

ICLN: iShares Global Clean Energy (solar, wind, renewables)

IXC: iShares Global Energy (oil, gas, integrated majors)

4 Jan 2021 – 31 Dec 2025 · 1,254 daily log returns each · no missing values

Yahoo Finance via yfinance, Close adjusted for dividends and splits

Daily log return	ICLN (clean)	IXC (traditional)
Standard deviation	1.81%	1.51%
Skewness	+0.20	−0.47
Excess kurtosis (Normal = 0)	2.04	2.67
Worst day	−7.62%	−8.84%
Best day	+8.02%	+6.82%

METHOD 1 · TAILS

Both ETFs have fat tails

19

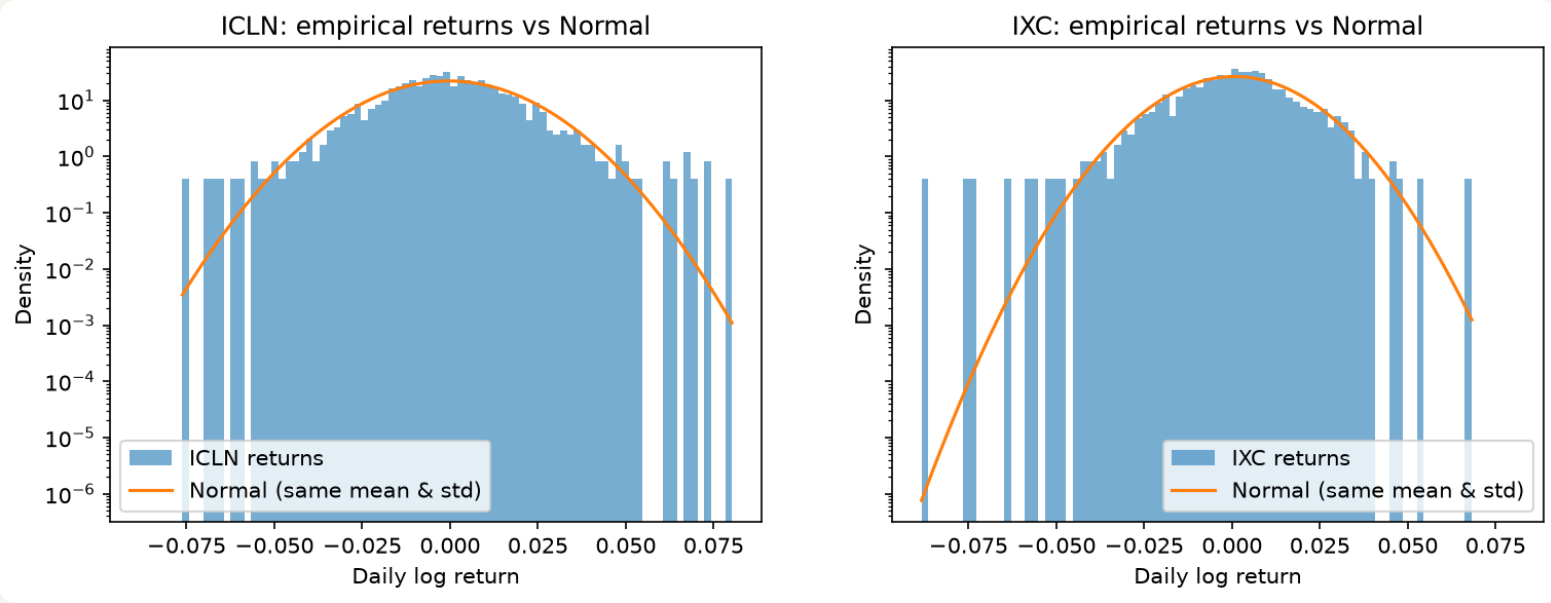
ICLN days beyond ±3σ

12

IXC days beyond ±3σ

3.4

days expected under a Normal in the same 1,254 days



Log scale: bars where the Normal curve is near 10^{-6} are days a Normal model treats as practically impossible.

METHOD 1 · NORMAL VS STUDENT'S T

Student's t fits far better than the Normal

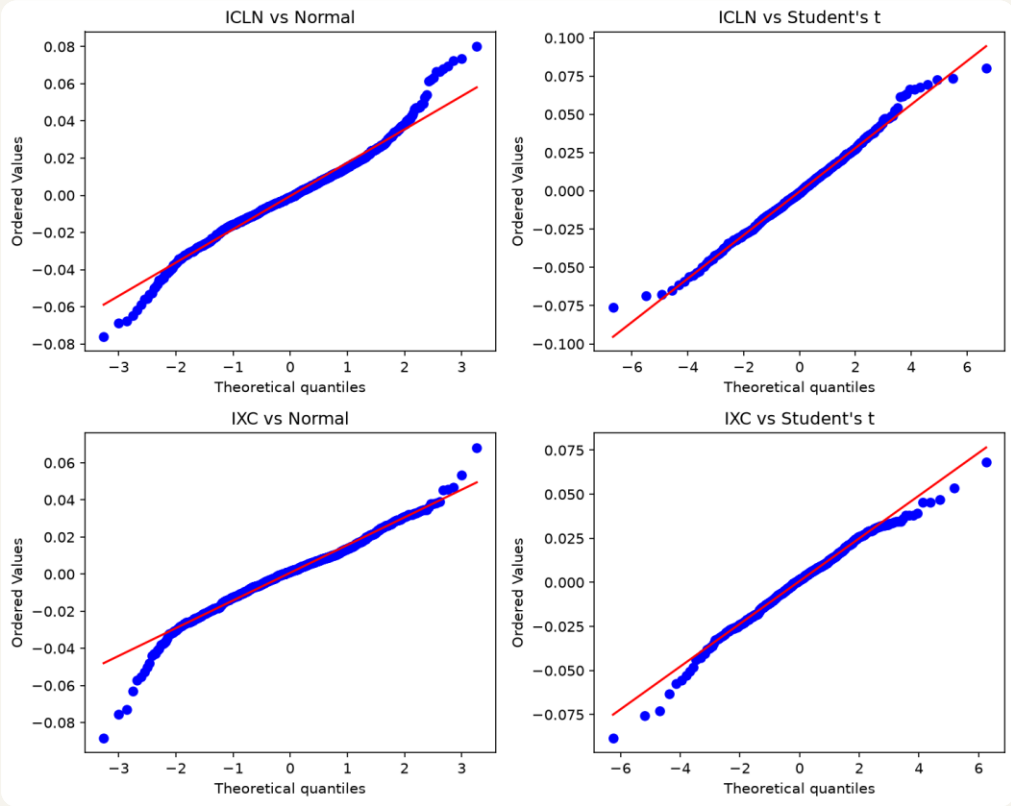
ν ≈ 5.0 and 5.5

Fitted degrees of freedom for ICLN and IXC. Small ν means fat tails; ν → ∞ is the Normal.

ΔAIC ≈ 87 and 84

In favour of the t, after penalising its extra parameter. Above 10 is strong evidence.

QQ plots: the Normal's S-shape shows tails too thin; the t straightens it, except IXC's two tails bend differently.



METHOD 1 · TAIL PROBABILITIES

The Normal misses extreme days; the t does not

Days beyond kσ	k	Actual	Normal	Student's t
ICLN	3	19	3.4	15.2
ICLN	4	4	0.08	4.6
ICLN	5	0	0.0007	1.7
IXC	3	12	3.4	14.0
IXC	4	5	0.08	4.0
IXC	5	2	0.0007	1.4

At 4σ the Normal expects **0.08 days** in five years; ICLN had 4 and IXC had 5, so it is **51× and 63× too low**. The t is within about a day.

METHOD 2 · CONCENTRATION INEQUALITIES

Two bounds that need no model

Markov

$$P(|r| \geq a) \leq E|r| \,/\, a$$

Needs a non-negative variable, so it is applied to |r|, the size of the move. Uses only the average move size (1st moment) and shrinks like 1/k.

Chebyshev

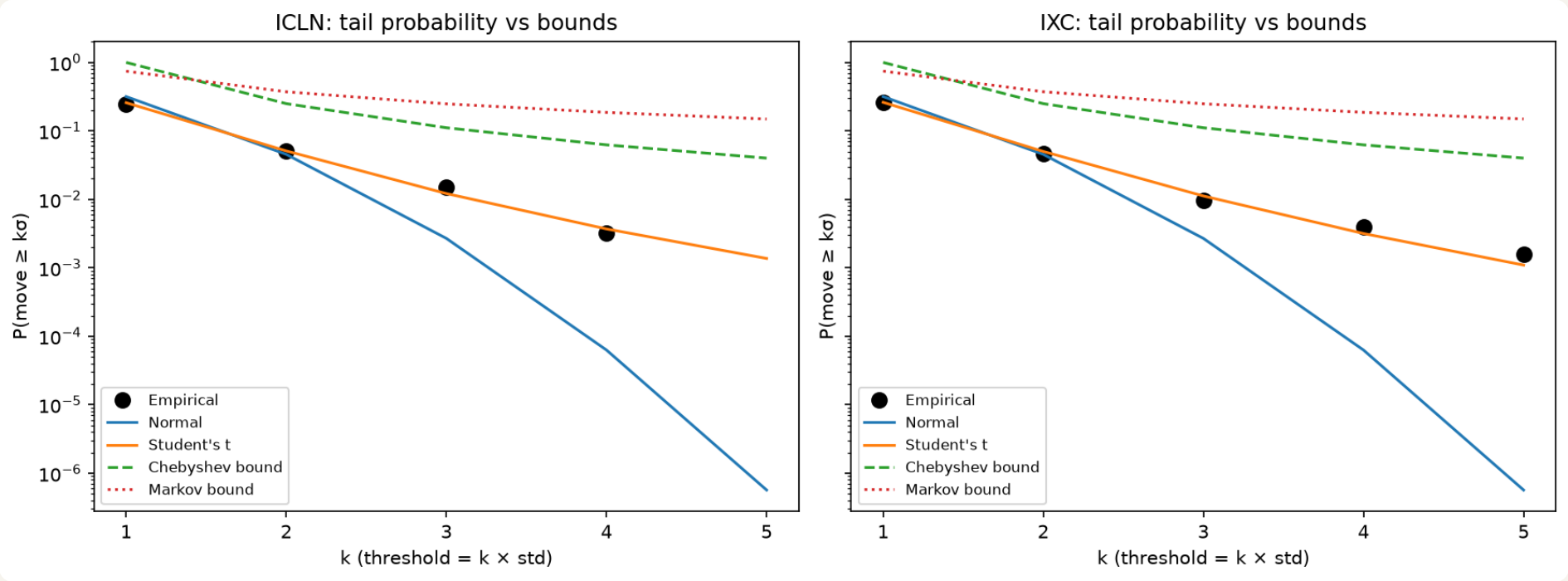
$$P(|r - \mu| \geq k\sigma) \leq 1 \,/\, k^2$$

Uses the variance (2nd moment) and shrinks like $1/k^2$. The same bound for every distribution, and so for both ETFs.

Chebyshev is Markov applied to $(r - \mu)^2$: more information gives a tighter bound.

METHOD 2 · BOUNDS VS REALITY

The bounds always hold, but overstate risk



Never violated (10 of 10 checks each). Chebyshev overstates by **4–25×**, Markov by **3–190×**, and the gap widens with k: real tails fall faster than $1/k^2$ and $1/k$.

WHAT A LOOSE BOUND MEANS IN PRACTICE

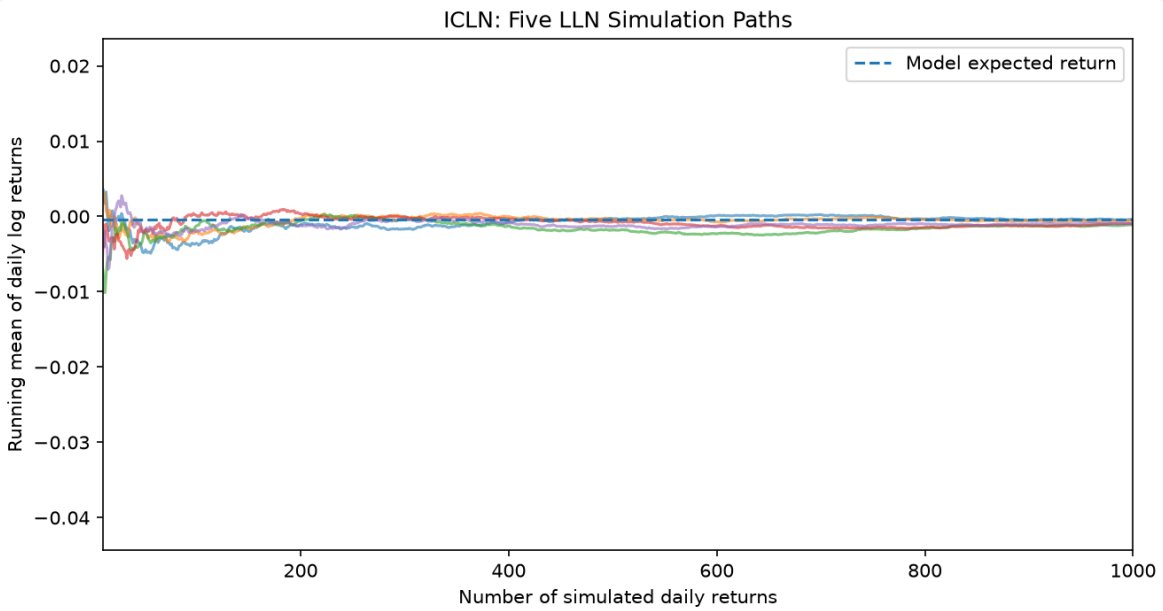
At 3σ, Chebyshev allows up to 11% of days.

Reality: 1.5% for ICLN and 1.0% for IXC.

Safe for any distribution, but a risk buffer sized from it would sit idle most of the time.

METHOD 3 · LAW OF LARGE NUMBERS

The sample mean converges, but slowly

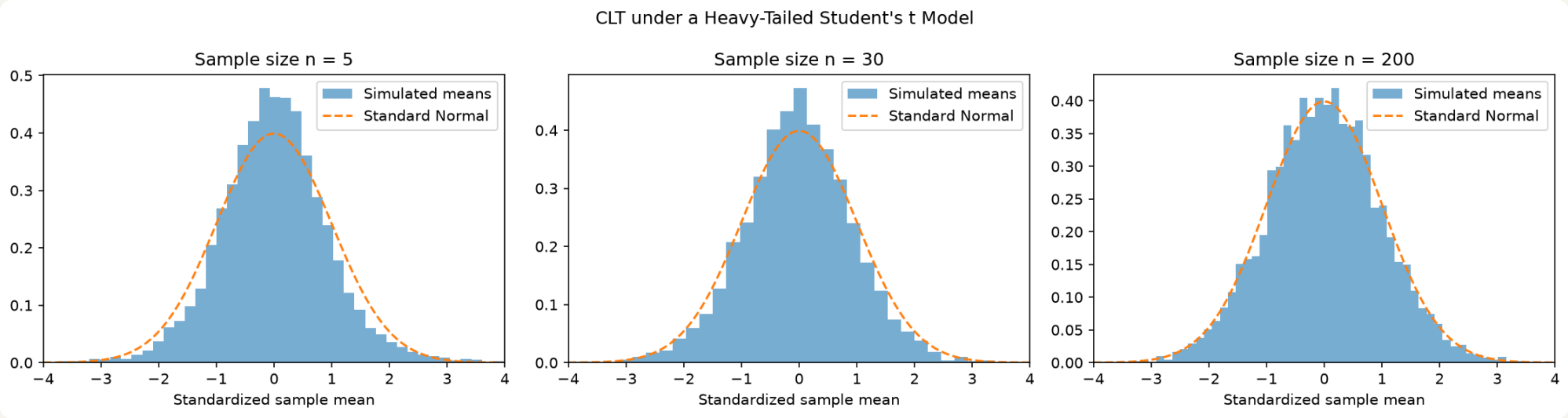


n	Miss > 0.2pp	SD of mean	σ/√n
10	74%	0.0057	0.0057
100	28%	0.0018	0.0018
1,000	0%	0.00058	0.00057

With our 1,254 real days, the uncertainty in ICLN's mean is about **0.051% a day**, larger than the mean itself (**−0.039%**).

METHOD 3 · CENTRAL LIMIT THEOREM

Averages of fat-tailed returns become Normal



n = 5

More peaked than Normal: peak ≈ 0.48 vs 0.40

n = 30

Closer, still slightly peaked in the middle

n = 200

Matches the standard Normal: the CLT has kicked in

CLEAN VS TRADITIONAL

Two sectors, two kinds of tail risk

ICLN · clean energy

- More volatile day to day: std 1.81%
- Slight positive skew (+0.20)
- 19 days beyond 3σ, none beyond 5σ
- Student’s t fits both tails well

IXC · traditional energy

- Calmer day to day: std 1.51%
- Negative skew (−0.47): crash-prone
- Worst day −8.84%; 2 days beyond 5σ
- Symmetric t misses its heavier left tail

Clean energy: more frequent moderate shocks. Traditional energy: rarer but larger crashes.

TAKEAWAY FOR A DECISION-MAKER

Which tool to trust for extreme days

NORMAL

Too optimistic

51–63× too few 4σ days. Fine for averages (CLT), not for single extreme days.

STUDENT'S T

Best fit

Within about a day of reality at 3–5σ. Misses IXC's crash asymmetry.

MARKOV · CHEBYSHEV

Too cautious

Always valid, but 3–190× too high. A worst-case check when nothing is known.

Use the t for tail risk, the bounds as a sanity ceiling, and the Normal only for averages.

LIMITATIONS

What these results cannot tell us

One five-year period

2021–2025 covers the clean-energy boom and bust and the 2022 energy shock; other periods may differ.

Independence assumed

All three methods treat days as independent, but real returns show volatility clustering.

Symmetric model, few extremes

The t cannot capture IXC's skew, and beyond 4σ there are only 0–5 days per ETF.

Simplified simulation

The CLT test used $\nu = 3$, not the fitted $\nu \approx 5$, and the LLN test used ICLN only.